

MSc project thesis

Parking Lot Allocation Strategy Design

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1 Introduction

This document is a graduation thesis for the Master thesis of Transport& Planning track of the Master of Civil Engineering in Delft University of Technology.

This research focuses on developing an optimal vehicle parking allocation strategy during Planned Special Events (PSE), such as football games or concerts, which involve a large influx of vehicles within a short time. Urban areas face numerous challenges, including congestion caused by factors such as the rapid increase in car ownership [15], parking space shortages, traffic accidents, etc. Planned special events contribute significantly to this congestion by generating high traffic volumes in a relatively small area within a limited time frame. Finding parking spaces for these vehicles quickly becomes crucial. While special events occur infrequently, they place substantial demands on traffic infrastructure, including parking facilities. Meeting this demand entails considerable financial costs. Unfortunately, parking space shortages are a common occurrence, particularly during peak hours[51]. Additionally, the time spent searching for parking spaces contributes to traffic congestion [59].

Large parking facilities can serve as valuable supporting infrastructure for planned special events, helping alleviate traffic pressure in the city and avoiding further traffic jams[39]. One way to relieve this burden is to optimize the operation of these parking facilities, letting the vehicles get parked as soon as possible, instead of waiting outside the parking lot or cruising on the road. Efficient operation policy on the parking lot is needed because the increasing number of cars also results in traffic congestion in parking lots[57]. Constructing more parking facilities to meet the demand during planned special events is not cost-effective. Instead, efficient parking operations can enhance the utilization of existing parking lots.

Numerous methods are employed to construct the parking system 'smart'. Sensors, GPS, and mathematical algorithms are used to sense the parking space availability[8], smart guidance[40], dynamic parking allocation [25], parking recommendation[17]. Besides, automation technology helps to enhance parking efficiency. Inside the parking lot, automated valet parking (AVP) saves human power and improves parking efficiency[59] [57] with full automation.

The search for parking spaces increases the risk of driver distractions and potential collisions [8]. Furthermore, environmental concerns, such as energy consumption, emissions, and noise, can be addressed by reducing unnecessary time spent searching for parking spaces[40][25]. Besides, organizing the parking process more effectively also leads to higher driver satisfaction[8].

During the planned special events, the demand for parking spaces is exceptionally high. Finding the optimal vehicle allocation strategy within the parking lot becomes crucial to improve organization and efficiency. First, it results in time savings and increased driver satisfaction by reducing the time spent searching for parking spaces. Second, it helps decrease potential congestion, thereby maximizing the utilization of the parking lot. By reducing turnover time, the parking lot can accommodate more vehicles within the same time period.

1 Introduction

This chapter serves as an introduction to this document. Chapter 2 outlines the objectives and research questions, Chapter 3 provides a literature review, Chapter 4 discusses the methodology used in this study, and Chapter 5 presents the experiment results.

2 Research Background

Urban traffic is facing significant pressure due to factors such as urbanization, increasing urban population, car ownership, and travel demand[33][21][2]. Traffic congestion has become a common and widespread phenomenon worldwide. The expansion of cities is accompanied by traffic congestion and its associated negative effects. Urban traffic congestion has detrimental impacts on both societal and personal benefits. These include long travel times, higher travel costs for travelers and shipping companies, air and noise pollution, high energy consumption, and greenhouse gas emissions[48]. Additionally, traffic congestion can lead to driver frustration and reduce the effectiveness of the traffic system [52][2]. It poses challenges to the sustainability and resilience of the traffic system [2][30]. Frequent braking during the congestion results in insufficient combustion of fuels and harmful emission gas[42][21][35]. Viard and Fu highlighted that traffic congestion contributes to air pollution through drivers' delays and tailpipe emissions. It also negatively affects economic growth by impacting labor costs, working hours, and delivery times[33]. The delays and costs associated with traffic congestion are particularly significant in densely populated areas[2].

Special events is one of the main cause of traffic congestion listed by the Federal Highway Administration (FHWA) [24]. On regular days, traffic demand can be considered as a repeated stochastic process with similar dynamic patterns within a day. However, during special events, traffic management needs to accommodate unusually high traffic demands [36]. Planned special events (PSE), such as football games, and concerts, is an important factor in urban congestion. PSEs are planned events that bring a substantial traffic in a short period, creating high traffic pressure in a concentrated area. The vehicles cruising in search of parking spaces exert considerable pressure on the surrounding road network and parking facilities. While constructing parking lots is a step towards addressing this issue, efficient operation strategies are equally important. Efficient operations enable vehicles to quickly find parking spaces, reducing cruising and waiting times. This, in turn, helps alleviate heavy traffic conditions.

Cruising for a parking lot increases the traffic pressure. Almost 30% of congestion is caused by cruising when a vehicle looking for a parking space[28]. During planned special events, drivers waste considerable time searching for parking spaces and waiting. Even though many cities provide a significant number of parking spaces on streets or in parking lots, drivers often spend a long time cruising or get stuck in congestion due to poor organization. While some cities offer information about available parking spots in certain areas, dynamic traffic conditions may render these spaces unavailable upon arrival, or drivers may spend excessive time searching for a spot within the parking lot. Large parking lots that can accommodate events like football games or concerts are crucial facilities for such venues[39]. However, the traffic within parking lots is complex and dynamic, and the high demand for parking spaces often leads to congestion without an efficient management strategy[57].

Developing an effective allocation strategy within parking lots that enhances safety and reliability can help reduce the time wasted in traffic cruising, alleviate traffic pressure, and

2 Research Background

maximize the utilization of parking infrastructure during planned special events. Additionally, it can contribute to energy savings and emission reduction in line with climate goals.

3 Objectives and Research Questions

3.1 Objectives

The objective of this project is to develop an optimal parking allocation strategy during planned special events to ensure efficient parking lot performance. The project workflow consists of the following components:

- Optimization problem formulation: Formulate the objective function, decision variables, and constraints for parking space allocation.
- Parking lot layout design: Determine a representative parking lot layout.
- Vehicle driving pattern recognition: Analyze how vehicles interact within the parking lot.
- Agent-based simulation construction: Develop an agent-based model to simulate the parking process realistically.
- Optimal strategy search algorithm: Design an efficient algorithm to search for the optimal parking allocation solution.
- Interpretation: Identify the characteristics and strengths of the optimal strategy.
- Validation: Apply the learned strategy to a simulated environment in NetLogo and compare it with control groups.

In summary, the study aims to achieve the following objectives:

- Formulate an optimization problem for parking allocation during busy periods.
- Develop an accurate agent-based simulation to model the parking process.
- Design an efficient algorithm to find the optimal parking allocation strategy.
- Validate the efficiency of the learned strategy through simulations.

3.2 Research Questions

Based on the objectives mentioned above, the main research question is:

What is the optimal strategy for efficiently allocating a large number of vehicles in a parking lot during highly busy times?

To address this main research question, the following sub-questions can be considered:

1. How should the optimization problem be formulated for parking allocation?

3 Objectives and Research Questions

- What is the objective function?
 - What are the decision variables?
 - What are the constraints?
2. How can the movement of vehicles be accurately modeled using agent-based modeling?
 - How do vehicles interact with other vehicles and with parking lot infrastructures?
 - What types of interference occur in the parking lot? How does the congestion occur and propagate in the parking lot?
 - How can the rules for the agent-based simulation be made more realistic?
 3. How should the algorithm be designed to find the optimal parking allocation strategy?
 - Is the algorithm efficient?
 - Does the algorithm produce an efficient solution?
 4. How should the learned strategy be interpreted, and what are its strengths and limitations?
 5. How can the efficiency of the solution be validated?
 - How should a simulation environment be constructed for validation?
 - What control strategies should be used for comparison?

By addressing these research questions, the study aims to contribute to the development of an optimal parking allocation strategy during planned special events.

4 Literature Review

4.1 Parking Management During Planned Special Events

Planned special events include sports events, concerts etc[20], which can deviate from the normal traffic distribution[36]. It has some characteristics: occurs at known locations and at scheduled times and the duration varies due to the event type [20]. The problem occurs within the temporal proximity to the event and spatial proximity to the event venue. So it is easier to organize and control than unexpected accidents. These events can cause severe traffic congestion and require effective traffic management strategies [20].

Researchers have proposed various approaches to manage parking during planned special events. For example, [20] proposed a mesoscopic simulation model to evaluate traffic impacts generated by PSE and assess mitigation measures reduced by traffic control plans. They used a dynamic traffic assignment (DTA) model to capture the temporal effects of congestion and the impacts of time-varying demands and supplies during the PSE. Tempelmeier et al. employed a supervised machine learning approach to predict the traffic impact of planned special events, came up with an algorithm to identify subgraphs of transportation graph that are typically affected by planned special events. Ruan et al. developed a space-time accessibility-maximization linear integer model to determine the scale and location of parking lots associated with mega-event sites. The goal was to maximize the number of travelers who could complete their tours within a reasonable travel time. Henao and Marshall conducted a study on parking at a sports event stadium in Denver, Colorado and found out that while the parking supply was sufficient, the parking provision was not efficient. They emphasized the need to consider the relationship between parking utilization and parking supply as part of an overall plan.

These studies highlight the importance of efficient parking management during planned special events to alleviate congestion and optimize parking utilization. Effective strategies, such as predictive modeling, optimization models, and comprehensive planning, can help ensure smooth traffic flow and enhance the overall event experience.

4.2 Parking Strategy inside Urban area

Efficient parking strategies within urban areas aim to reduce the time spent searching for parking spaces, alleviate traffic congestion, and enhance the overall parking experience. Advanced technologies and algorithms have been proposed to make parking systems "smart" and optimize parking allocation.

Researchers have developed various solutions for smart parking systems. Promy and Islam came up with an Android-based navigation system to help drivers to find a parking space nearby. To reduce the traffic volume in urban areas caused by vehicles searching for parking,

Geng and Cassandras introduced a smart parking system to allocate and reserve the optimal parking spaces for drivers using a mixed-integer linear programming (MILP) problem. Wu et al. proposed ParkHop, a mobile crowdsensing system that aggregates the availability of parking spaces both on the street and in the parking lot through sensorless sensing and disseminates this information with up-to-date prices to drivers. Rahman et al. developed a mobile deep learning method to predict parking occupancy to reduce drivers' searching time for parking spaces. Zhao et al. developed a dynamic parking distribution algorithm considering shared and non-shared modes to maximize parking resource utilization in the city. Xie et al. created a parking allocation model considering the ability of dynamic and static traffic conversion and the solution method is based on particle swarm optimization, which balanced the utilization of surrounding parking resources and the reduction of the dynamic traffic pressure. Nakazato et al. proposed a smart-parking system with a reservation system and a dynamic parking fee design.

Internet of Things (IoT) technology gives some intriguing solutions for urban parking space allocation. Arellano-Verdejo et al. developed a mathematical model and evolutionary algorithm to optimize the allocation of public parking spots in a smart city. Wireless Sensor Networks have been used to create parking lot guidance systems, allowing drivers to find nearby parking spaces through mobile apps. GPS traces and sensors have been utilized to sense on-street parking space availability, improving traffic efficiency. Shin et al. came up with a smart parking guidance algorithm for city transportation management, considering the driving distance, walking distance, and traffic congestion. Anusha and Pushpalatha proposed a smart parking guidance algorithm that considers driving and walking distances, as well as traffic congestion. Sangeetha et al. use IoT to assist parking systems to address traffic congestion, parking spaces lack, and safety issues.

Autonomy of the traffic system can contribute to parking efficiency. Sayarshad designed intelligent public parking allocation for autonomous vehicles, allowing AVs to move from crowded city centers to less congested areas for parking. A dynamic optimization formulation is proposed considering drivers' preferences including rent bid, parking lot searching time, and travel costs, which is solved by a Lagrangian relaxation algorithm.

These research efforts demonstrate the potential of smart technologies, IoT, and autonomous vehicles in improving parking allocation and reducing traffic congestion within urban areas. By leveraging these advancements, it is possible to enhance parking efficiency and alleviate the stress of finding parking.

4.3 Vehicle Allocation Strategy Within a Parking Lot

The advancement of smart technologies and algorithms has enabled the implementation of intelligent parking systems within parking lots. These systems utilize controllers, sensors, computer vision, and IoT technology to optimize vehicle allocation and enhance the overall parking experience.

Smart parking systems incorporate various technologies to assist drivers in finding parking spaces. Asaduzzaman et al. proposed a ZigBee-based smart parking system, a wireless communication protocol, for heavy traffic environments. It can hold multi-floor buildings and send messages to vehicles about the status of parking spaces. Prabagar et al. used computer vision to monitor the availability of spots and the entry and exits of the vehicles. Bibi et al. presented an automatic smart parking system based on computer vision to assist

drivers to find a suitable parking space and to reserve it. [Athira et al.](#) used optical character recognition to identify the availability of parking spaces. IoT technology enables accessibility to things from a remote location. [Mahendra et al.](#) implemented IoT-based sensors in the parking system to help drivers reserve parking spaces from a remote location. [Huang et al.](#) integrated parking reservations into automated valet parking (AVP) system, in which a privacy-preserving reservation scheme was proposed. [Patil et al.](#) demonstrated a centralized parking system for drivers to reach the free parking slots through the fastest route.

Dynamic control and planning methods are commonly employed within parking lots due to the dynamic nature of parking space availability. Real-time GPS data also helps in dynamic parking allocation. [Mladenović et al.](#) set up a dynamic parking allocation scenario using GPS data to guide the vehicle to the designated parking lot solving a 0-1 programming model. [Shin et al.](#) developed a dynamic control of intelligent parking guidance using neural network predictive control to operate an intelligent parking guidance system, which is for assessing and selecting the best parking lot in a real-time environment. [Duan et al.](#) solving a 0-1 programming model. [Fransen et al.](#) studied Real-time path planning for large, dense grid-based automated guided vehicle (AGV) systems using a dynamic approach. [Yang et al.](#) analyzed parking space utilization and developed a dynamic model for parking space allocation to improve driver satisfaction and occupancy balance.

The use of automated guided vehicles (AGVs) within parking lots has gained attention for enhancing efficiency and reducing congestion. Researchers have developed deep reinforcement learning algorithms, such as Double Dueling Deep Q-Network and DDPG, to optimize parking space allocation in AVP systems. These approaches consider multi-objective optimization, balancing individual and cooperative allocation effects, and incorporating rewards for global efficiency and model learning. [Zhang et al.](#) proposed a cooperative approach for multi-AGV systems inspired by hierarchical traffic control, solving parking space allocation using deep reinforcement learning (DRL). Cooperative driving is also implemented at the lower level to avoid congestion and collisions.[11]. While reinforcement learning is tolerated for the reward delay between the environment and generalized planning. The allocation of parking spaces consists of the action set of this system. They propose a Double Dueling Deep Q-Network based on the original deep Q-network(DQN) for the DRL. [Xie et al.](#) used a system-side DRL-based cooperative approach with a global objective for parking space allocation, considering immediate and future effects. Their algorithm, based on deep deterministic policy gradient (DDPG), improves global efficiency and collision avoidance. Reinforcement learning-based approaches have been employed for parking space allocation. [Zhang et al.](#) developed an online parking assignment strategy in an environment with partially connected vehicles using a multi-agent deep reinforcement learning approach.

Overall, the research in vehicle allocation strategies within parking lots focuses on leveraging smart technologies, computer vision, IoT, and autonomous vehicles to optimize parking space allocation, improve efficiency, and enhance the parking experience.

4.4 Parking Traffic Model

Choice behaviors affect the model formulation, especially in an agent-based model. Every vehicle should make its own decisions such as choosing a route, searching for parking space, staying in the parking space for a while, and finally exiting, etc [47]. [Vo et al.](#) divided

parking movement into 2 steps: route choice for parking strip and route choice for parking space, which are highly affected by choice behaviors. Factors such as travel time to the parking space, walking time to the destination, and guidance signs affect drivers' choice of parking space. Empirical analysis is often used to construct parking choice behavior and statistical models for arrival and departure. Attributes like travel time, walking distance, and parking probability are determined to study drivers' preferences. [Xu and Sun](#) looked into the "arrival priority" and "reservation priority" in parking management and designed an "ant colony-genetic" algorithm to solve it.

[Xue et al.](#) proposed that shared parking can be used to alleviate urban parking problems and came up with an integer programming model with continuous time windows based on global information. Hierarchical architecture models, such as bi-level simulation optimization systems, are commonly used for parking space allocation. [\[57\]\[47\]\[13\]](#). Aiming to find the optimal solution for parking space management, a bi-level simulation optimization system is constructed [\[47\]](#) and found out that different parking time limitations help to enhance the parking lot efficiency. The upper level is an optimization model aimed at maximizing the social benefit running in Matlab, including driving time, searching time, and walking time. The operation management policies are simulated in the lower level, which is a multi-agent-based simulation based on a NetLogo environment. The output of the lower level is used back for tuning the upper-level model.

[Dogaroglu et al.](#) compare an Intelligent Parking Guidance System (IPGS) model with a conventional system with regard to capacity utilization. A multi-agent-based model simulated using NetLogo demonstrated improved detection of available parking spaces and reduced searching time.

In a multi-region urban road network, macroscopic modeling for autonomous vehicles (AVs) with dynamic control[\[59\]](#) is used to reduce cruising-for-parking time. The optimization of the distribution of floating AVs and route guidance is achieved by a centralized parking dispatch approach.

Microscopic models, such as VISSIM software, are employed for traffic simulation within parking lots[\[39\]](#). It is a rather complex evacuation problem considering a large amount of audience attending the games of the 2008 Beijing Olympic Games at the National Stadium in China. Simulation steps are structure data entry, vehicle data entry, existing traffic controlling mode and virtual detectors setting, flow data entry, simulation, and result analysis. The result shows that the simulation in VISSIM is highly accurate for the validation of the evacuation plan. Microscopic simulation has some advantages[\[39\]](#): it provides a more precise representation compared to macroscopic simulation, considering factors like passenger diversity, vehicle features, driving skills, highway network, and organizing efforts.[Makarova et al.](#) utilized microscopic simulation to evaluate the efficiency and additional effects of urban parking organization.

Overall, parking traffic models encompass choice behaviors, empirical analysis, shared parking solutions, hierarchical architecture models, macroscopic and microscopic simulations to optimize parking space allocation, improve capacity utilization, reduce cruising time, and enhance efficiency in parking systems.

4.5 Research Gap

One research gap in the field of parking lot allocation is the lack of studies specifically addressing efficient allocation during large events, such as football games, which typically have high traffic volumes. While there has been research on parking lot allocation in urban areas and single parking lots, there is a scarcity of studies that specifically focus on the unique challenges and requirements of efficiently allocating parking spaces during large events.

Large events pose specific challenges due to the sudden influx of vehicles and the need to accommodate a significant number of attendees. Efficient parking lot allocation during these events requires strategies that can handle the high demand and optimize the utilization of parking spaces to ensure smooth traffic flow and minimize congestion.

Therefore, there is a need for research that specifically addresses the efficient allocation of parking spaces during large events. This research could explore innovative approaches, algorithms, and optimization techniques to handle the unique characteristics and challenges associated with high traffic volume and time-constrained parking requirements.

By addressing this research gap, it is beneficial to the development of practical and effective strategies for parking lot allocation during PSEs, ultimately improving traffic management and enhancing the overall experience for event attendees.

5 Methodology

The methodology for addressing the parking lot allocation problem involves formulating it as an optimization problem and designing an algorithm to find the optimal strategy for maximizing benefits. The methodology includes mathematical problem formulation, agent-based simulation, solution searching, and validation.

- **Mathematical Problem Formulation:** The parking lot allocation problem is defined as an optimization problem, where the objective is to maximize benefits. The allocation strategies are considered as solution points within the solution space.
- **Agent-Based Simulation:** An agent-based simulation is constructed to simulate the parking process of vehicles within the parking lot. This simulation environment captures the interactions between vehicles and infrastructures, providing detailed information for decision-making.
- **Solution Searching:** Several algorithms are employed to search for the optimal strategy. Initially, standard linear optimization programming is used, but since it lacks the ability to consider vehicle interactions, a genetic algorithm is utilized in combination with the simulation to find the optimal solution. Additionally, reinforcement learning, a machine learning technique, is tested in the experiment to train agents in an environment to maximize rewards and improve efficiency.
- **Validation:** To validate the effectiveness of the learned strategy, a set of control group strategies is defined. The efficiency of these control group strategies is compared against the optimal strategy obtained from the solution searching process. This validation module evaluates and verifies the effectiveness of the optimal strategy.

The overall research process involves applying these steps: mathematical problem formulation, agent-based simulation, solution searching, and validation. The agent-based simulation module provides a realistic environment for solution searching, while the validation module assesses the performance of the optimal strategy compared to the control group strategies. It is shown in figure [5.1](#).

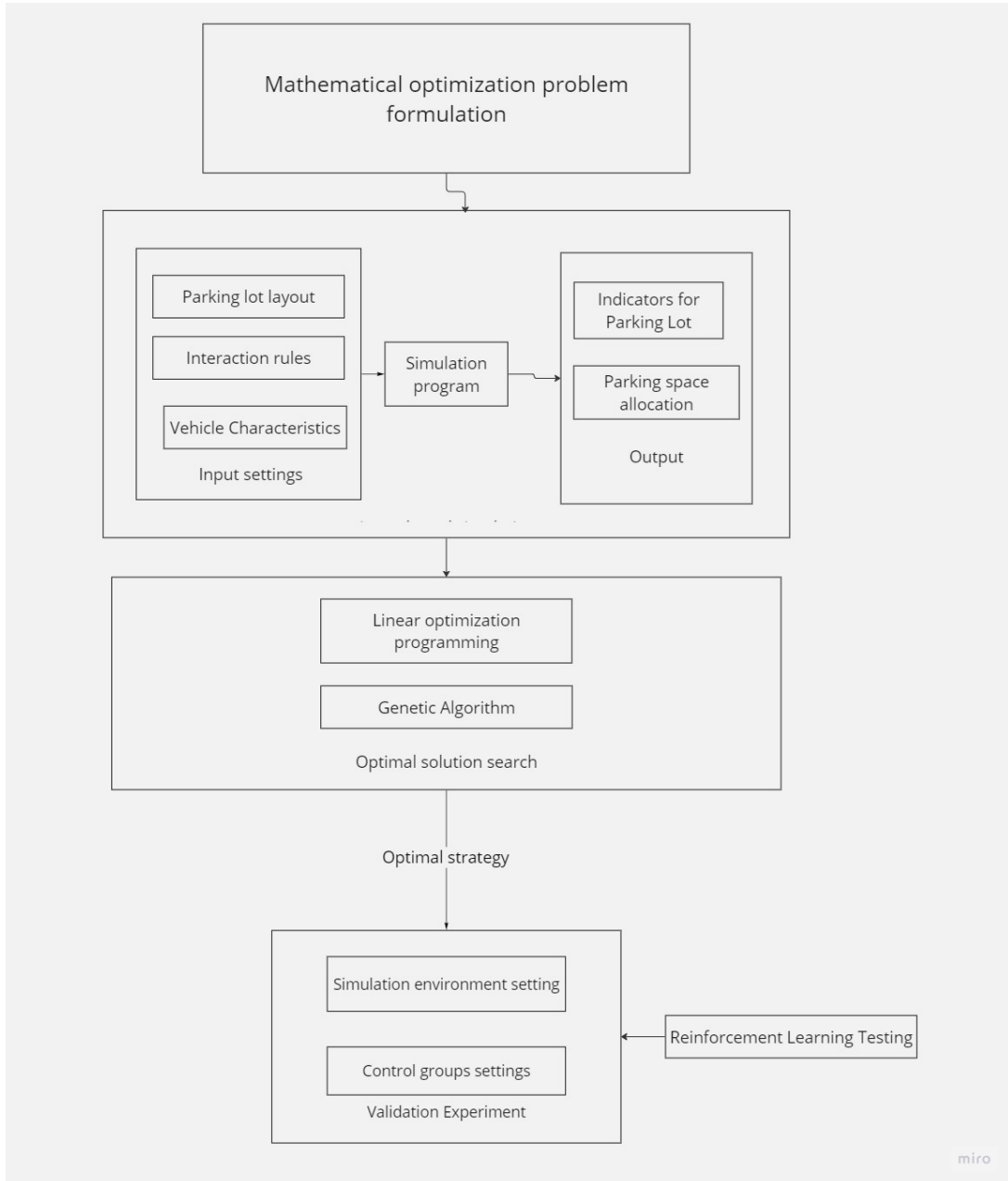


Figure 5.1: Research Process

5.1 Optimization Problem Formulation

There are N parking spaces in the parking lot in total. As a result, it can accommodate at most N vehicles.

Considering the benefit for all vehicles, the objective function is to minimize the sum of travel time for all vehicles.

$$\min z = \sum_{i=1}^N t_i, \quad (5.1)$$

t_i , travel time for the vehicle i .

For constraints, every vehicle should be allocated a parking space. Besides, every parking space can only contain one vehicle.

$$\sum_{j=1}^N x_{ij} = 1, \quad i \in 1, 2, 3, \dots, N \quad (5.2)$$

$$\sum_{i=1}^N y_{ij} = 1, \quad j \in 1, 2, 3, \dots, N \quad (5.3)$$

x_{ij} , a binary variable that is 1 if vehicle i is assigned to parking space j , and 0 otherwise.

y_{ij} , a binary variable that is 1 if parking space j is assigned to parking space i , and 0 otherwise.

5.2 Agent-based simulation module

5.2.1 Introduction

Agent-based modeling, also known as multi-agent-based modeling, is commonly used to simulate travel behavior, including parking behavior[47][58]. For example, an agent-based simulation is used for on-street parking space searching [10] and drivers' movement inside the parking lot[47]. It allows for the explicit modeling of individual drivers' behavior and their interactions, providing insights into their movement patterns.

NetLogo is open-source software for multi-agent-based modeling simulation showing advantages for behavioral simulation[47], which can be used to simulate travel behavior for a large number of agents and allow inputs like non-homogeneous environment and personal preferences[59]. It has its own programming language and is suitable for a small system. Users can create their own models in NetLogo with the simple scripting language and user-friendly interface. NetLogo allows researchers not only to look into the microscopic level considering individual behaviors of drivers but also the macroscopic level coming from their interactions[47].

In this section, a agent-based simulation implemented in NetLogo allows for a microscopic analysis of individual driver behaviors. By simulating the behavior of each driver agent, insights can be gained into the parking process at both the individual and collective levels, providing a realistic representation of the parking lot dynamics. Each vehicle in the parking lot can act as an agent to obey some simple rules. The simulation shows the entering and searching of the vehicles. The exiting is not considered here.

5.2.2 Parking Lot Layout

The parking lot layout used in the agent-based simulation is based on *Layout2* shown in figure 5.6 which is derived from a layout in a microscopic simulation shown in figure 5.2[47]. It focuses on the vehicle movement in the parking lot and also used simulation software NetLogo, with the same focus and the same simulation software as this research. It is designed to resemble a real parking lot in Eindhoven[47], ensuring a realistic representation of the parking environment. The layout only includes the entrance, parking spaces and lanes of the parking lot and excludes other infrastructures. The simulation specifically focuses on the entering behavior of vehicles and does not consider the exiting process.



Figure 5.2: Parking Simulation Interface[47]

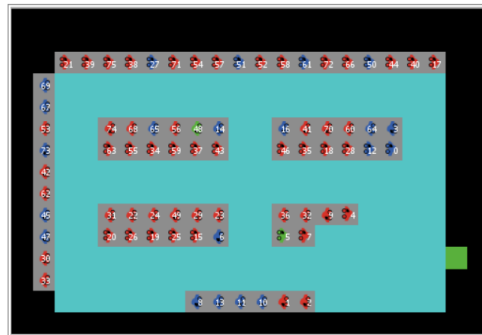


Figure 5.3: Filled Parking Lot Layout 2

The movement inside the parking lot is complex. They can change lanes, and parking on the right or left. So it is easy to have some collisions or congestion. To simplify the understanding of vehicle movement within the parking lot, a preliminary layout called *layout1* is created(Figure 5.4). In this layout, vehicles can only be parked on the right side, reducing the number of parking spaces and providing fewer route choices for vehicles. Additionally, some roads are designed with only one lane to avoid conflicts and facilitate a step-by-step understanding of the movement dynamics.

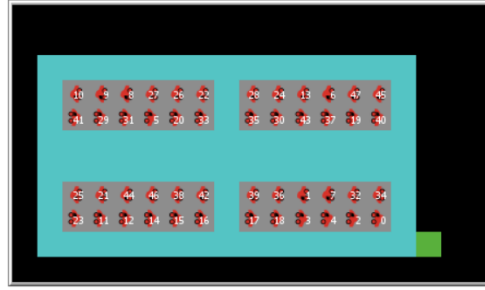


Figure 5.4: Directions of lanes in layout 1

Within the simpler parking layout *Layout1*, specific rules govern vehicle movements. When vehicles drive in a straight lane, a car following model is used to maintain appropriate spacing. At intersections, vehicles change their direction without any speed loss to keep the simulation simplified. When a vehicle decides to choose a parking space, it temporarily blocks the lane as it requires time to park. This behavior results in reasonable movement patterns within the parking lot. After it finishes the parking process, the congestion gradually resolves.

To introduce more dynamics, the second parking layout *Layout2* is investigated. Vehicles in *Layout2* have the option to park on either the right or left side. Since vehicles drive on the right side of the road, if they want to park on the left, they must change lanes. Lane changes are only allowed when the left side is empty. Once the lane change is completed, the process of parking into the designated space remains the same as in *Layout1*. Figure 5.5 illustrates the directions of lanes in *Layout1*, showing the flow of traffic.

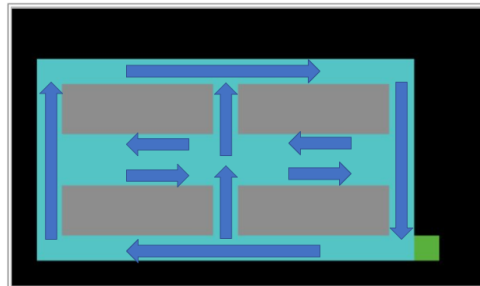


Figure 5.5: Directions of lanes in layout 1

Layout2 has a similar size to *Layout1*, but it allows vehicles to change lanes when they spot an open parking space across the street. This enables vehicles to select parking spots on both sides. Additionally, the space close to the entrance is larger compared to a single road layout.

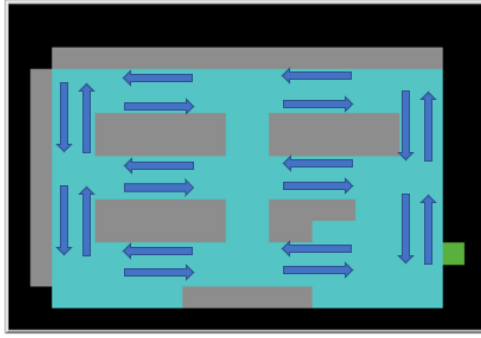


Figure 5.6: Directions of lanes in layout 2

5.2.3 Basic parameters setting

Patches are squares that make up the world of the model in NetLogo[[noa](#)]. Each patch corresponds to a 5.5 m by 5.5 m square parking space, aligning with the recommendation of the Urban Plan Institute[[58](#)][[47](#)]. The following parameters are set for the simulation, as shown in Table [5.1](#).

Table 5.1: Parameters setting

v	speed limit	30 km/h[58]
$L_{headway}$	expected car following distance	4.0 m [43]
u_{max}	max acceleration	$5m/s^2$ [43]
d_{max}	max deceleration	$-2.5m/s^2$ [43]
$L_{vehicle}$	length of vehicle	3.72 m [19]

Parameters were set through the scroll bar at the simulation interface. The interface for the simulation constructed in NetLogo is shown in figure [5.7](#).

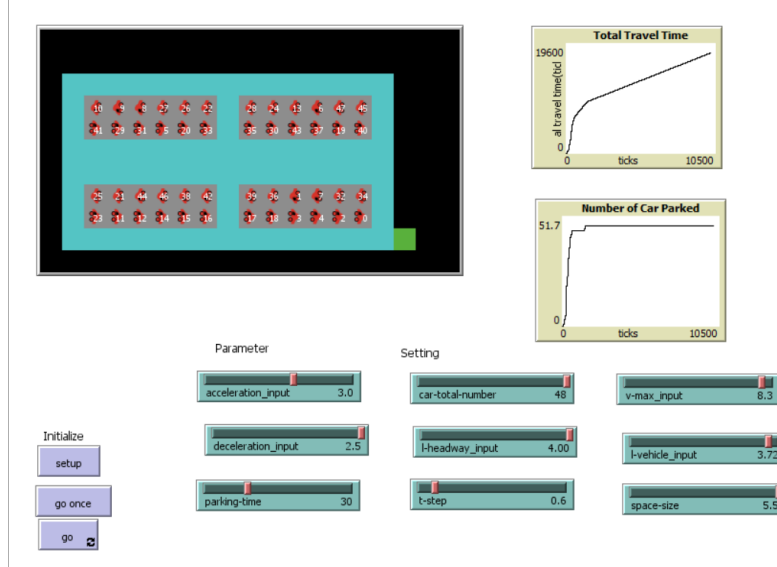


Figure 5.7: Simulation Interface

5.2.4 Agents Interaction Rules

Vehicles cannot enter the parking lot altogether. They have to enter the parking lot one by one and keep a distance. After entering the parking lot, there are interferences inside the parking lot. Like the seat interferences and aisle interferences in the airline boarding[?], the interference can exit on the road or at the parking spaces in the parking lot. The following Vehicles have to decelerate or stop when blocked by the ego vehicles. When a vehicle is parking in a spot, it can also interfere with other surrounding vehicles.

A few assumptions are made for agents' interaction rules:

- All agents are identical and follow the same rules.
- All parking spaces are identical, and the time required to park is the same regardless of the availability of spaces on both sides.
- All agents arrive at the parking lot simultaneously, entering one by one. The time spent waiting outside the parking lot is included in their travel time.

Car-to-car Interaction Rule

When vehicles are on straight lanes, the Newell's Car-Following Model is used.

Newell's car-following model is a simple time-discrete model for traffic flow modeling. The speed function is given by the free flow speed, gap, and time step.

$$v(t + T) = \max[0, \min(v_0, (s - s_0/T))] \quad (5.4)$$

$v(t)$, speed function with time variable t

T , time step

v_0 , maximum speed

s , distance gap between 2 vehicles

s_0 minimum distance gap between 2 vehicles

Car-to-Environment Interaction Rules

The rules for vehicles to interact with the environment are: entering the parking lot, parking into the parking space, making turns, or keeping the original direction when at the crossing.

- Entering the parking lot
Each vehicle will arrive at the entrance when both the patch before it and the entrance are vacant. The entering process continues until all parking spaces are filled.
- Parking into the parking spaces
The routes to each parking space are predetermined before the simulation starts, eliminating the freedom for vehicles to choose a route. Each vehicle takes time t_{park} to park once it reaches its allocated space.
- Making turns or maintaining direction at crossings
Vehicles change their direction or maintain their original direction when at a crossing, without any speed loss.

The simulation continues until all vehicles are parked, at which point the simulation stops.

5.3 Optimization Algorithm

5.3.1 Linear Optimization Algorithm

In order to solve the optimization problem described in section 5.1, a traditional linear optimization programming approach is employed. The estimation of travel time is a key component of this algorithm, which comprises both the driving time and waiting time. Initially, it is assumed that the driving time is linearly related to the driving distance and is represented by the equation:

$$t_i = \alpha d_{ij} + tw_i \quad (5.5)$$

where:

t_i , travel time for vehicle i

d_{ij} , distance for vehicle i to the allocated space j

tw_i , the waiting time for vehicle i

However, it should be noted that the waiting time for vehicles is influenced by the order in which they enter the parking lot, as a vehicle may block the following vehicles, leading to non-linear waiting times. To address this, a linear formulation is adopted, assuming that the travel time is solely dependent on the distance:

$$t_i = \alpha d_{ij} \quad (5.6)$$

5.3.2 Genetic Algorithm

The Genetic Algorithm (GA) is an optimization algorithm inspired by natural selection and genetics. In this approach, candidate solutions are represented as arrays known as chromosomes. As the algorithm iterates through generations, solutions with higher fitness are retained as parents for the subsequent generation, while lower fitness solutions are discarded. The process of generating offspring involves recombination through crossover and mutation, while also reparation considering constraints.

Chromosome Coding

When route choice is not available and the order of vehicles is followed, a parking space is assigned to each vehicle. The allocation solution can be represented as a chromosome, where each gene represents a parking space with a vehicle assigned. As shown in figure 5.8, *chromosome_1* means that the first parking space is assigned to vehicle 3, the second is assigned to vehicle 5, and so on.

3	5	1	4	...	...
1	2	3	4	...	N

Figure 5.8: Chromosome 1 Coding For Parking Space Allocation

In situations where vehicles have the option to choose between parking on the left or right, an additional chromosome *chromosome_2* is defined to represent the direction in which the vehicle will drive to the assigned parking space. As shown in figure 5.9, 1 indicates that the assigned parking space is on the right of the vehicle, while 0 represents that it is on the left.

0	1	0	0	...	...
1	2	3	4	...	N

Figure 5.9: Chromosome 2 Coding For Parking Space Allocation

Crossover

The crossover process involves generating new offspring by recombining the genes of two parents. Two parents are randomly selected from the parent population, and gene exchange occurs at a designated crossover location.

For N is an even number in this case, the crossover location is defined as the middle point of the chromosomes. The resulting offspring is formed by combining the first half of parent 1 with the second half of parent 2, as shown in figure 5.10.

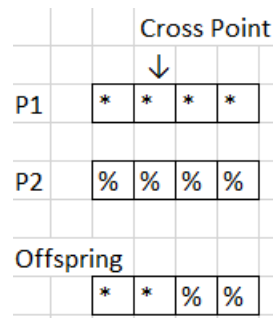


Figure 5.10: Crossover

Mutation

Mutation is an important aspect of exploring the solution space. Genes have a mutation rate, allowing them to mutate into another randomly chosen gene. Figure 5.11 depicts this mutation process.

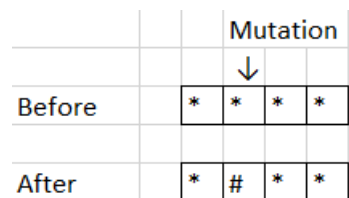


Figure 5.11: Mutation

Reparation

To account for the constraints present in the optimization problem, *chromosome_1* requires all genes to be unique since each vehicle can only be allocated to one parking space. After the crossover and mutation operations, every gene is examined. If a gene is found to be non-unique within the chromosome, it is replaced by another randomly available gene until uniqueness is achieved.

Selection Strategy

Individuals with higher fitness are kept as the parents of the next generation. The fitness is scaled with the value of the objective function z . Since the objective is to minimize z , the individuals with lower z have higher fitness.

The value of fitness is calculated like the following:

- For the population with M individuals, calculate the objective value z for each individual. Then the maximum z_{max} among the population is determined.
- The fitness function f for each individual is defined as a function of z :

$$f(z) = z_{max} - z \quad (5.7)$$

After the calculation for fitness, individuals with fitness in the top 50% are selected as parents for generating the offspring. The number of offspring equals the number of parents, and these parents and offspring form the population for the next generation.

Hyperparameters

When employing the Genetic Algorithm for searching the optimal solution, several hyperparameters need to be considered:

population size, the number of individuals in the population

num_generations, the number of iterations or generations of the algorithm

mutation rate, probability for mutation during the algorithm

5.4 Validation module

For the validation of different strategies, the simulation is conducted using NetLogo. Two different layouts, *Layout1* and *Layout2*, are considered.

5.4.1 Layout 1

The parking spaces are labeled as shown in 5.12:

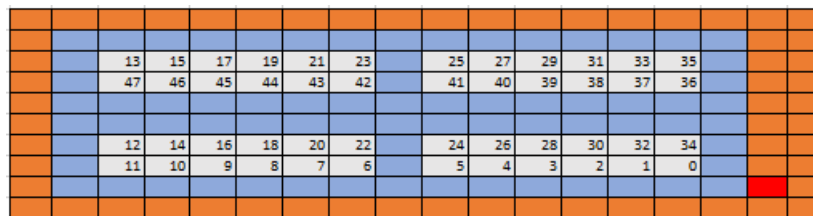


Figure 5.12: Layout 1 Label

The parking spaces are divided into several zones based on the routes depicted in Figure 5.13.

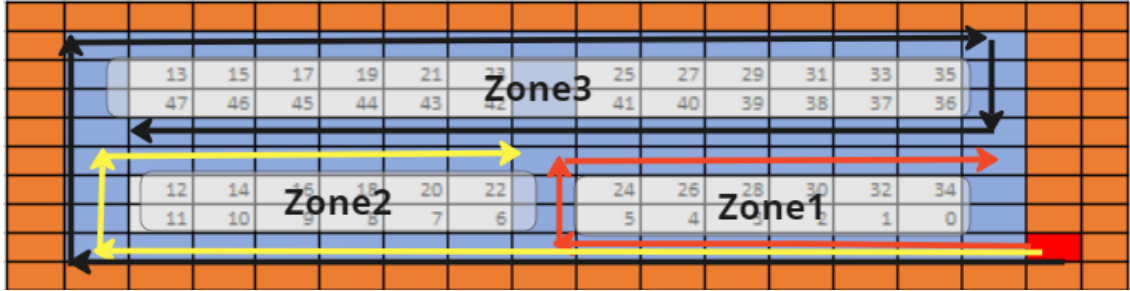


Figure 5.13: Routes and Zones Layout 1

To compare different strategies, the following allocations are defined:

1. The parking lot is filled along routes from the nearest to the furthest, and vehicles can only select the parking space on the right
2. The parking lot is filled from the furthest to the nearest, and vehicles can only select the parking space on the right

5.4.2 Layout 2

In *Layout2*, similar to *Layout1*, the parking spaces are labeled, as shown in Figure 5.14:

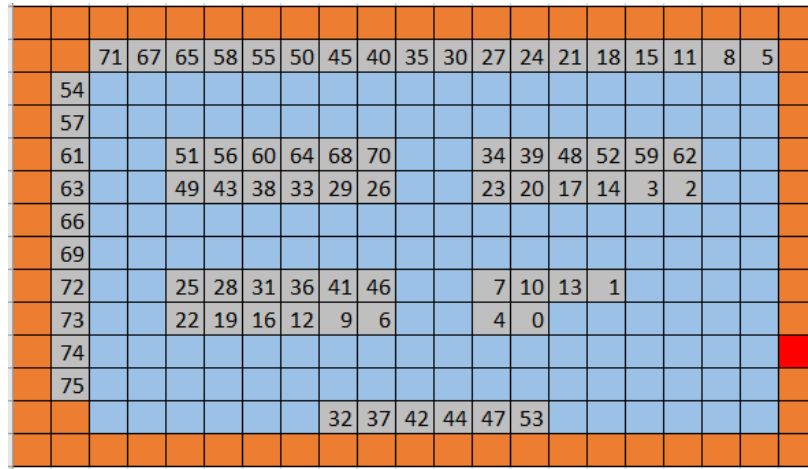


Figure 5.14: Layout2 Parking Lot Label

Due to the grid layout and constraints in road directions, the route choice within the parking lot is limited. The road directions are indicated in Figure 5.15.

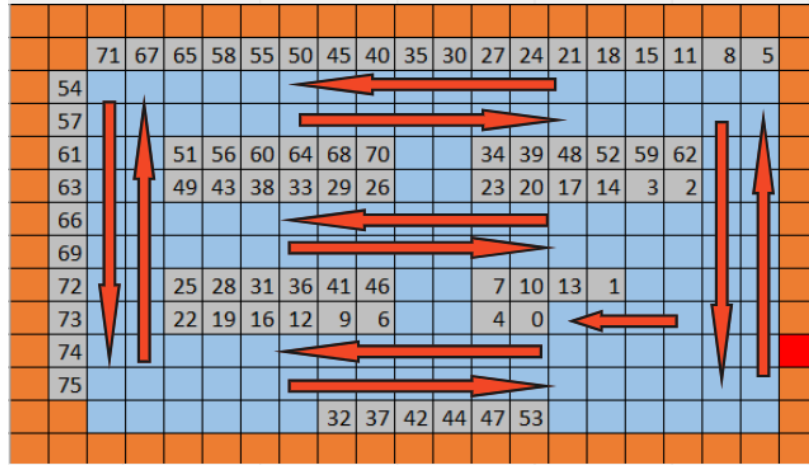


Figure 5.15: Road Directions *Layout2*

The route choice in the parking lot is quite limited due to the grid layout and constrain in the road direction. Letting each vehicle get to the allocated parking space by the shortest route, 4 comparison strategy is defined:

1. The parking lot is filled along routes from the nearest to the furthest, and vehicles can only select the parking space on the right
2. The parking lot is filled from the furthest to the nearest, and vehicles can only select the parking space on the right
3. The parking lot is filled from the nearest to the furthest, and vehicles can only select the parking space on the left
4. The parking lot is filled from the furthest to the nearest, and vehicles can only select the parking space on the left

6 Experiment

6.1 Comparison Between Linear Optimization Algorithm and Genetic Algorithm

In this section, we present a comparison between the Linear Optimization Algorithm and the Genetic Algorithm (GA) using a simple example.

The example consists of a parking lot layout with 12 parking spaces, as shown in Figure 6.1:

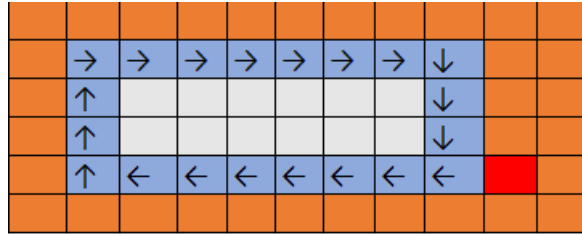


Figure 6.1: A Simple Parking Lot Layout

The vehicles can only park on the right side, and the route is indicated by arrows.

Focusing on the process inside the parking lot, distance is the route length from the entrance to the allocated parking space, which means that it is independent of the vehicles themselves, but only relevant to the parking space location. It is noticed that there is only one route for vehicles to drive to their allocated parking space. Since all vehicles are assigned a parking space and all spaces are occupied, the sum of travel distances for all vehicles is a constant:

$$\sum_{i=1}^N d_{ij} = c \quad (6.1)$$

c , a constant

As a result, the objective function is also a constant, which means that all the solutions have the same objective value:

$$z = \sum_{i=1}^N t_{ij} = \sum_{i=1}^N d_{ij} = c \quad (6.2)$$

However, this scenario is unrealistic because interference inside the parking lot, such as vehicle blocking and congestion, is not considered.

6 Experiment

To address these issues, we apply the Genetic Algorithm in combination with simulation. The parameters for GA are set as follows:

population size, 100

num_generations, 100

mutation rate, 0.5

The simulation layout in NetLogo is shown in figure 6.2:

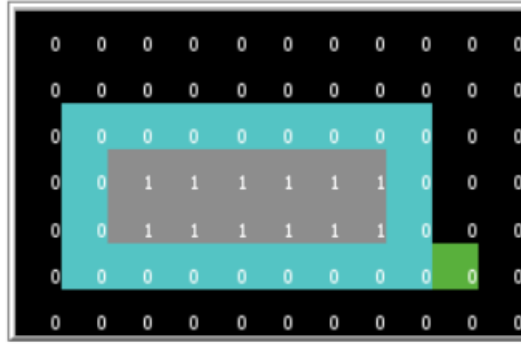


Figure 6.2: Simulation layout in NetLogo

The objective values across generations are shown in Figure 6.3.

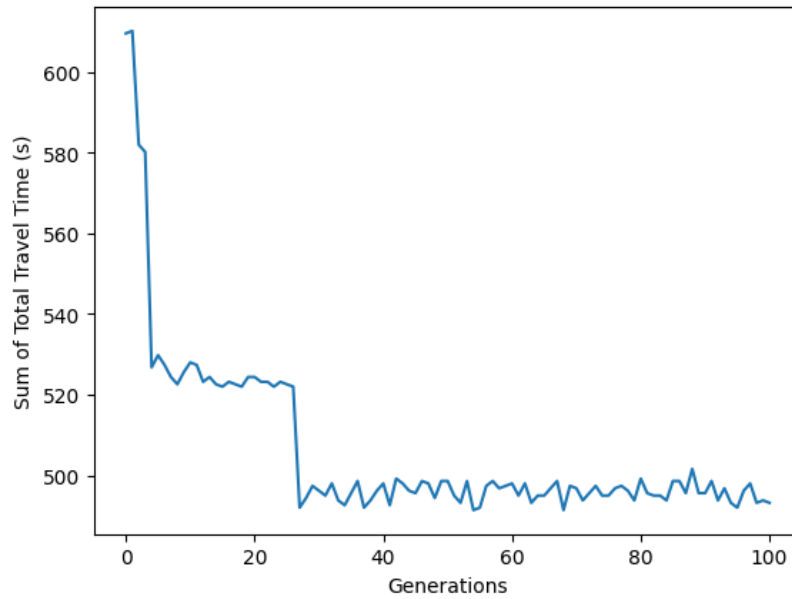


Figure 6.3: Objective Values Across Generations For the Simple Layout

6 Experiment

The objective value initially decreases and converges after approximately 20 generations. The solution found by GA is:

[8 1 7 0 6 5 11 2 9 10 3 4]

The optimal strategy allocation in the parking lot is shown in figure 6.4:

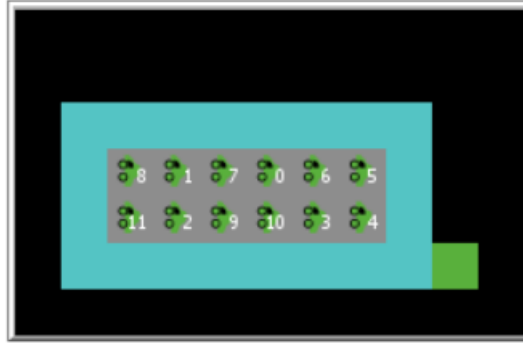


Figure 6.4: Optimal Strategy Allocation For the Simple Layout

To compare this solution with another strategy, we consider a strategy where each vehicle chooses the nearest parking space upon entering. The allocation for this comparison strategy is presented in Figure 6.5:

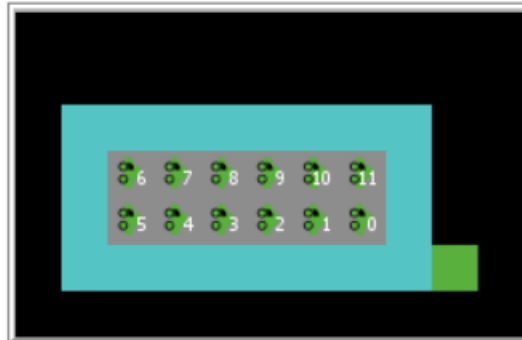


Figure 6.5: Comparison Strategy Allocation For the Simple Layout

The total travel time for each strategy is shown in figure 6.6, where can be seen clearly that the optimal strategy we got used less time and less total travel time to let all vehicle park.

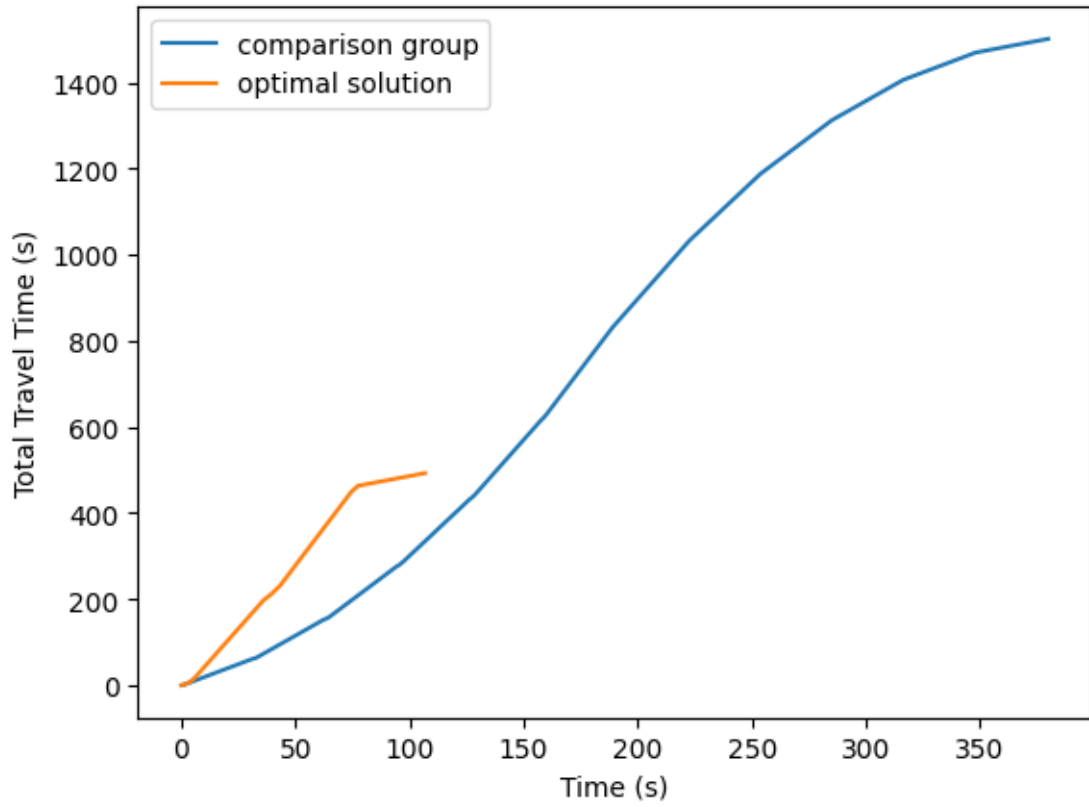


Figure 6.6: Total Travel Time For the Simple Layout

The solution obtained through GA exhibits better performance and is more realistic than the linear optimization algorithm in this case. Based on these results, GA is selected for the solution search in the subsequent sections.

6.2 Layout 1

The evolution of the objective value across generations for Layout 1 is presented in Figure 6.7:

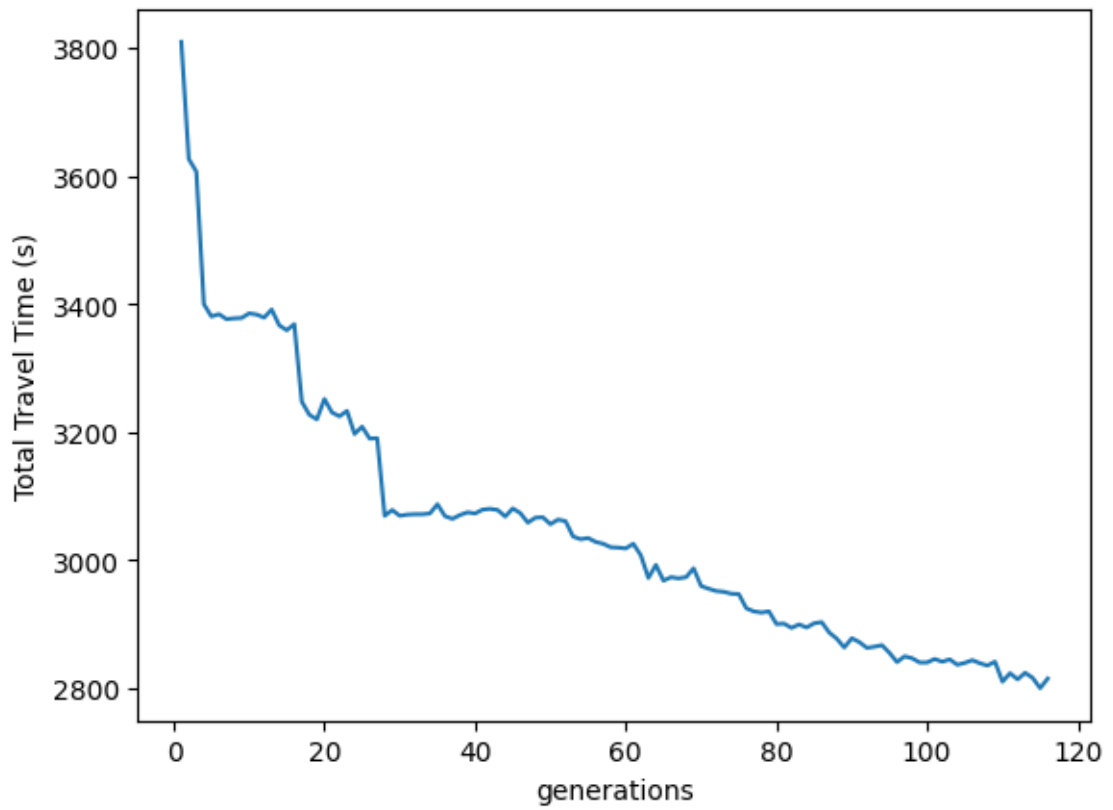
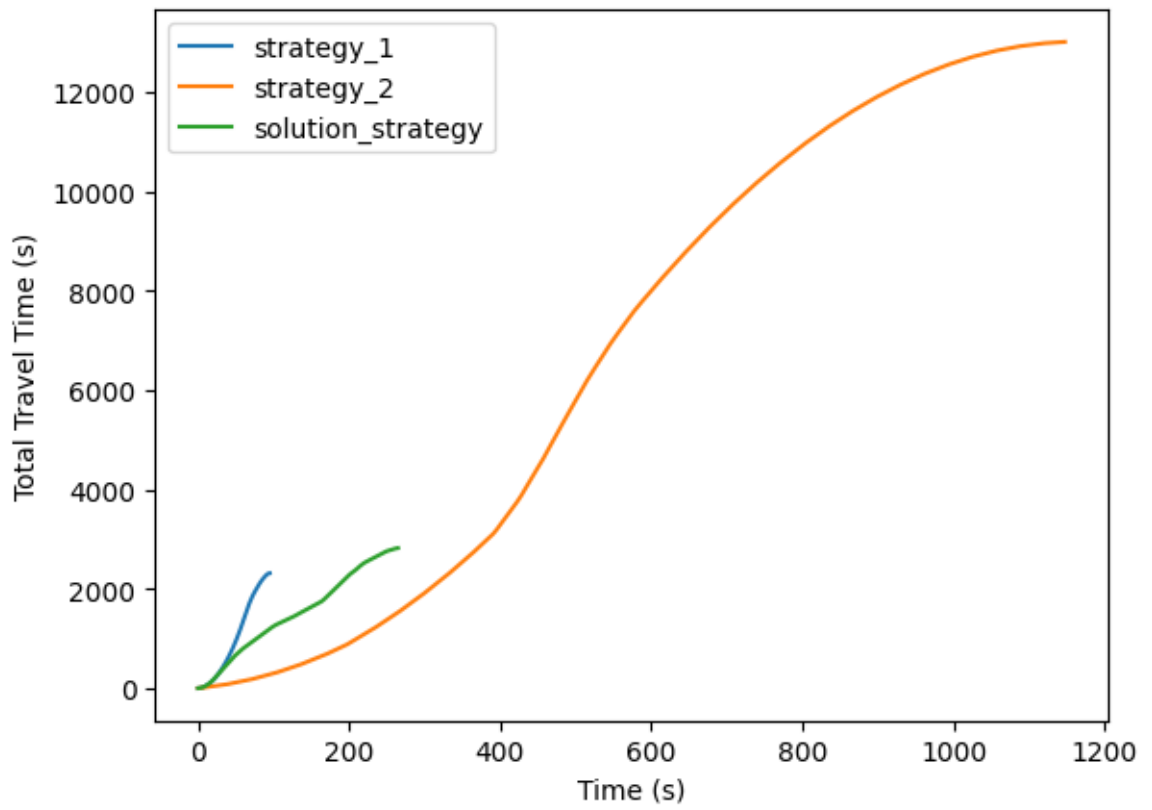


Figure 6.7: Total Travel Time Across Generations *Layout1*

The total travel time during the parking process for the solution and comparison group is displayed in Figure 6.8:

Figure 6.8: Total Travel Time Parking Process *Layout1*

A steady improvement can be seen as the GA progresses. This indicates that the algorithm is effectively searching for better solutions and optimizing the total travel time. The total travel time during the parking process further confirms the superiority of the GA solution over the comparison group. The GA-generated solution achieves a significantly lower total travel time, indicating more efficient parking allocation and reduced congestion within the parking lot.

6.3 Layout 2

The solution for allocation is shown in figure 6.9. The red vehicles are parked on the right side, and the blue vehicles are parked on the left side.

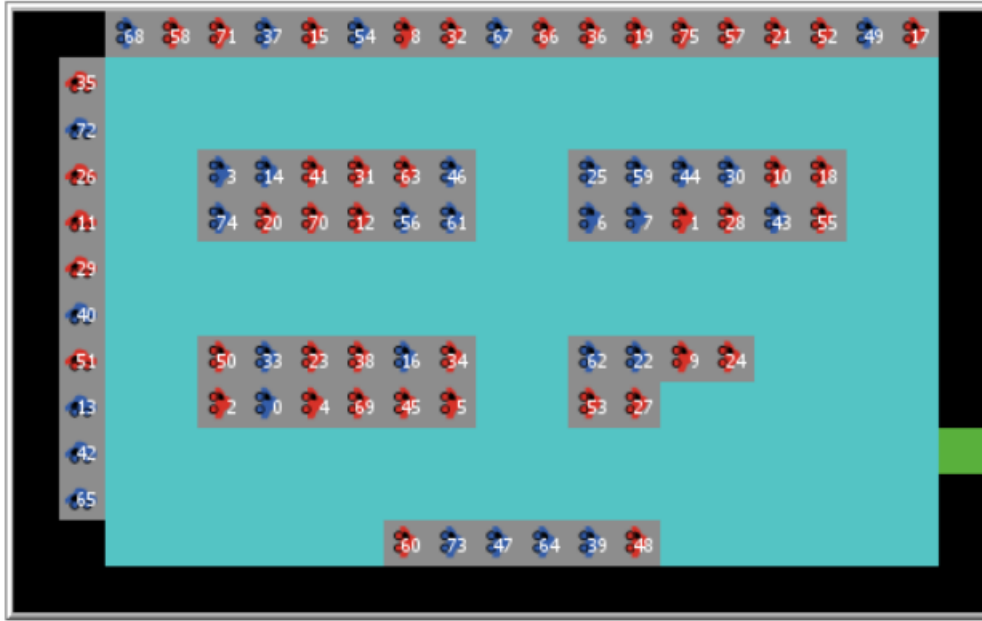


Figure 6.9: Solution *Layout2*

The evolution of the objective value across generations for *Layout2* is shown in Figure 6.10:

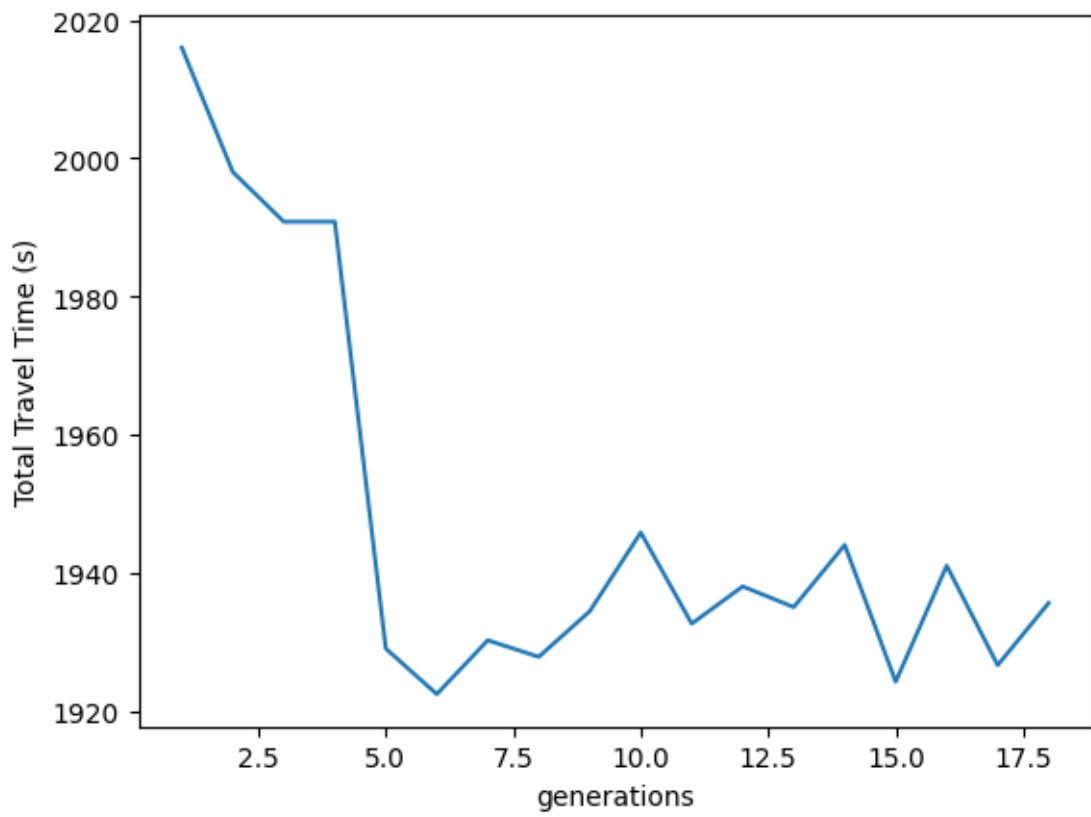
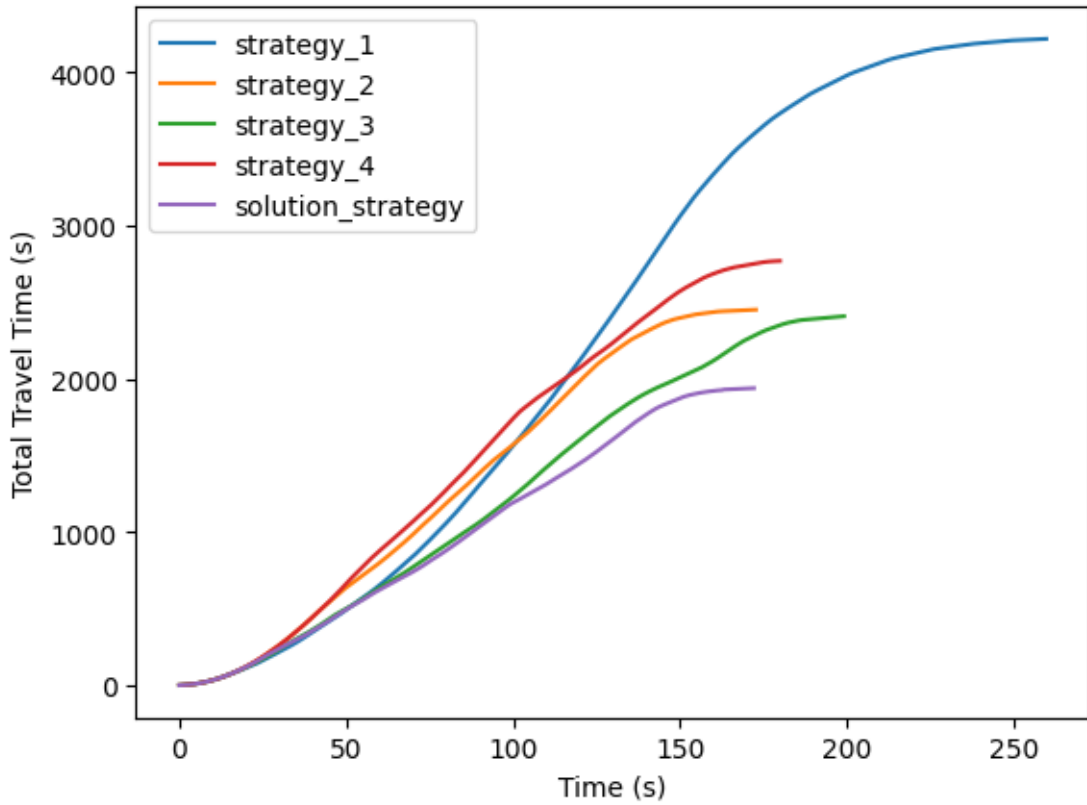


Figure 6.10: Total Travel Time Across Generations *Layout2*

The total travel time during the parking process for the solution and comparison group is illustrated in Figure 6.11:

Figure 6.11: Total Travel Time Parking Process *Layout2*

The solution obtained through GA demonstrates the lowest total travel time in this case.

In *Layout2*, vehicles have the option to choose between parking on the left or right side, the solution found by the GA, showcases the effectiveness of the algorithm in handling complex scenarios. The GA evolves the objective value across generations, leading to a solution with the lowest total travel time. This observation is reinforced by comparing the total travel time during the parking process for the GA solution and the comparison group. Once again, the GA outperforms the comparison group, highlighting its ability to find optimized solutions in layouts with greater route choice flexibility.

6.4 Discussion

Based on the experiments conducted on the two layouts, it can be observed that the Genetic Algorithm performs better when dealing with more complex layouts that offer more route options. While it may be relatively easy to find a good solution for simpler layouts with a limited number of routes, the Genetic Algorithm proves to be an effective approach for finding optimal solutions in more complex scenarios.

These results emphasize the advantage of employing the GA approach for parking lot optimization, particularly in scenarios where the layout is complex and offers multiple route options. The GA's ability to explore and evaluate various combinations of parking space allocations enables it to discover solutions that minimize travel time. By leveraging the power of genetic operators such as crossover, mutation, and selection, the GA efficiently navigates the solution space, converging towards improved solutions over generations.

It is worth noting that while the GA demonstrates promising performance across these experiments. Additionally, the fine-tuning of hyperparameters, such as population size, number of generations, and mutation rate, could lead to even more optimal results.

In conclusion, the Genetic Algorithm proves to be a valuable tool for optimizing parking lot allocation, particularly in layouts with complex configurations. Its ability to adapt and evolve solutions over generations makes it well-suited for addressing the challenges of parking lot optimization, improving efficiency, and reducing congestion.

6.5 Reinforcement Learning

Reinforcement learning (RL) is also used in parking space allocation. In this section, it is tested for the efficiency in the optimal solution searching.

When a vehicle enters the parking lot, the system will allocate a parking space for it according to the parking lot state, with 3 elements of perception, action, and goal, which can be modeled as a Markov decision process so that reinforcement learning can be applied[51] [44]. In this case, the agent is the system, with the perception of the parking lot, and conducts an action to allocate a parking space.

The RL environment is based on the simulation defined in 6.5.3, and then is defined as a Gym custom environment. Gym is a Python library for conducting research in reinforcement learning. The training framework is built with the help of Pytorch, which are compatible with the Gym ecosystem. After the game environment and the training framework construction, the training process is applied to get the learned strategy.

6.5.1 Introduction

The standard process for reinforcement learning can be described as Markov decision process[9]. What is discussed here is discrete states with finite action spaces.

A finite Markov decision process is defined as:

$$\langle X, U, f, \rho \rangle \quad (6.3)$$

X	<i>finite environment state set</i>
U	<i>finite agent action set</i>
$f : X \times U \times X \rightarrow [0,1]$	<i>the state transition probability function</i>
$\rho : X \times U \times X \rightarrow \mathbb{R}$	<i>reward function</i>

The process evolves as the discrete time steps go by. At time step k , the state is $x_k \in X$. The agent gets an observation from the state x_k and chooses an action $u_k \in U$. After that, the state changes to $x_{k+1} \in X$ at time step $k+1$ according to the transition probabilities f , which is ending up as $f(x_k, u_k, x_{k+1})$. The agent gets a reward $r_{k+1} \in \mathbb{R}$ according to the reward function $\rho : r_{k+1} = \rho(x_k, u_k, x_{k+1})$, which reflects the immediate effect of the action. The reward function is assumed bounded.

Agents choose their actions according to their observation of the state. The behavior of the agents is described by its policy. The policy can be stochastic: $h : X \times U \rightarrow [0, 1]$ or deterministic policy $\bar{h} : X \rightarrow U$

The agents' goal is to maximize the expected discounted return for every state:

$$R^h(x) = E\left\{\sum_{k=0}^{\infty} \gamma^k r_{k+1} | x_0 = x, h\right\} \quad (6.4)$$

In this case, $\gamma \in [0, 1)$ is the discount factor, reflecting the long-term effect of an action. The return R represents the accumulative reward for action in the long run.

6.5.2 Algorithms

Q Learning

To maximize the long-term return with only receiving the immediate reward, the optimal state-action value function (Q-function) for the single agent is defined: $Q_h : X \times U \rightarrow \mathbb{R}$, which gives the expected return obtained by the policy h from any state-action pair:

$$Q^h(x, u) = E\left\{\sum_{k=0}^{\infty} \gamma^k r_{k+1} | x_0 = x, u_0 = u, h\right\} \quad (6.5)$$

Similarly, the Q-function for the multi-agent case is defined for each agent depending on the joint action and the joint policy, $Q_i^h : X \times \mathbf{U} \rightarrow \mathbb{R}$:

$$Q_i^h(x, \mathbf{u}) = E\left\{\sum_{k=0}^{\infty} \gamma^k r_{i,k+1} | x_0 = x, \mathbf{u}_0 = \mathbf{u}, h\right\} \quad (6.6)$$

The optimal Q-function is defined as:

$$Q^*(x, u) = \max_h Q^h(x, u) \quad (6.7)$$

$Q^*(x, u)$ satisfies the Bellman optimality equation:

$$Q^*(x, u) = \sum_{x' \in X} f(x, u, x') [\rho(x, u, x') + \gamma \max_{u'} Q^*(x', u')] \quad \forall x \in X, u \in U \quad (6.8)$$

It means that the optimal Q value for taking u in x equals to the immediate reward plus the expected optimal value attainable from the next state.

Once we got the Q^* , the optimal policy can be computed by choosing the action in every state with the highest optimal Q-value:

$$\bar{h}^*(x) = \arg \max_u Q^*(x, u) \quad (6.9)$$

If there are multiple actions that can attain the highest Q-value, any of them can be chosen and the policy remains optimal. In such a case, it only returns one of these equally good solutions. A policy that maximizes the Q-function value in this way is called greedy. As a result, an optimal policy can be found by determining Q^* and then computing a greedy policy in Q^* .

Q-learning is a model-free algorithm[49]. It uses an iterative approximation procedure to solve this problem. Q-learning starts with an arbitrary Q-function, observing transitions $(x_k, u_k, x_{k+1}, r_{k+1})$. After each transition, the Q-function is updated with:

$$Q(x_k, u_k)_{k+1} = Q_k(x_k, u_k) + \alpha_k [r_{k+1} + \gamma \max_{u'} Q_k^*(x_{k+1}, u') - Q_k(x_k, u_k)] \quad (6.10)$$

$\alpha_k \in (0, 1]$ is the learning rate. The term between the square brackets is the difference between the current estimate $Q_k(x_k, u_k)$ and the updated estimator $r_{k+1} + \gamma \max_{u'} Q_k^*(x_{k+1}, u')$.

Deep Q-Network(DQN)

DQN combines reinforcement learning with a class of artificial neural networks, which is known as deep neural networks, to learn policies directly from high-dimensional inputs[26]. It is based on Q-Learning.

The approximate value function is parameterized as $Q(s, a; \theta_i)$ using deep convolutional neural network. θ_i are the parameters(weights) of the Q-network at iteration i . Agent's experiences $e_t = (s_t, a_t, r_t, s_{t+1})$ at each time step t are stored in a data set $D_t = e_1, \dots, e_t$ to perform experience replay. The samples of experience (s, a, r, s') $U(D)$ are applied with Q-learning updates at each iteration i during training. The Q-learning update at iteration i uses the following loss function:

$$L_i(\theta_i) = \mathbb{E}_{(s,a,r,s') \sim U(D)} [(r + \gamma \max_{a'} Q(s', a'; \theta_i^-) - Q(s, a; \theta_i))^2] \quad (6.11)$$

- γ discount factor determining the agent's horizon
- θ_i parameters of Q-network at iteration i
- θ_i^- network parameters for computing target at iteration i , which are updated with θ_i every C steps

The action-value function is estimated by a function approximator:

$$Q(x, u; \theta) \approx Q^*(x, u) \quad (6.12)$$

In DQN a non-linear function approximator is used, as the neural network. It is called Q-network, which is a neural network with weights θ . A Q-network is trained by adjusting the

parameters θ_i at iteration i to reduce the MSE(mean-squared error) in equation 6.8, where the optimal target value $r + \gamma \max_{u'} Q^*(s', a')$ are substituted with approximate target values $y = r + \gamma \max_{u'} Q^*(s', a'; \theta_i^-)$, the parameters θ_i^- .

The loss function iterates:

$$L_i(\theta_i) = \mathbb{E}_{s,a,r}[(\mathbb{E}[y|s,a] - Q(s,a;\theta_i))^2] = \mathbb{E}_{s,a,r,s'}[(y - Q(s,a;\theta_i))^2] + \mathbb{E}_{s,a,r}[\mathbb{V}_{s'}[y]] \quad (6.13)$$

The targets depend on the weights of the Q-network. Parameters θ_i are fixed when optimizing the loss function $L_i(\theta_i)$. The final term is the variance of the targets and is independent with the parameters θ_i , so it can be ignored. Differentiating the loss function with respect to θ_i :

$$\nabla_{\theta_i} L(\theta_i) = \mathbb{E}_{s,a,r,s'}[(r + \gamma \max_{a'} Q(s', a'; \theta_i^-) - Q(s,a;\theta_i)) \nabla_{\theta_i} Q(s,a;\theta_i)] \quad (6.14)$$

6.5.3 Gym Custom Environment Creation

The game is defined based on the simulation constructed in 6.5.3 as a Gym custom environment. Gym is an efficient Python library for conducting research reinforcement learning. Gym custom environment can be tailored easily to meet the special need. The structures are very clear.

Making a custom Gym environment out of all the parts makes the outcome replicable. Other packages that are compatible with the Gym environment can also be used.

Environment and Agents

The agent is the system which assign the parking spaces to vehicles. The total number of vehicles is N .

The environment is the parking lot abstracted from the simulation in , which is an array containing the information about the available parking spaces. It is 1 if the parking space is occupied and 0 if not. The shape of array is $1, N$. For example,

$[1, 0, 1, 1, 0, \dots, 0]$

Observation and Action Space

It is assumed that the environment is fully observed. That is to say, observation equals the state of the environment.

The action set consists of the labels of the available parking spaces. Labels are integer numbers. For example, here is an action set:

$[5, 8, 13, 16]$

It means that there are 4 available parking spaces in the parking lot, and their labels are 5, 8, 13, and 16, respectively.

Reward Function

The reward function focuses on system optimal, which means vehicles should spend as little as possible time on the parking process. Variable t_travel_i is defined as the time from the first vehicle enters the parking lot until the vehicle i has parked.

The time spent by all vehicles in the parking lot:

$$\min.z = \sum_{i=0}^{N-1} t_travel_i \quad (6.15)$$

Which is equal to:

$$\max.z = - \sum_{i=0}^{N-1} t_travel_i \quad (6.16)$$

A problem here is the delay between the action and the reward associated with the travel time. To overcome the delayed feedback, when every M vehicles parked, system gets a reward, which is linear to the travel time. So the total reward is linear to the sum of total travel time of all vehicles. The reward is scaled within 100. Reward r is defined:

$$r = \begin{cases} (-avg_M + 100)/(100 - 35) * 100 & \text{if there are } M \text{ more vehicles parked} \\ 0 & \text{else} \end{cases}$$

avg_M , the average travel time for M vehicles.

If M is too low, it will tend to park the vehicle early, which will block the following vehicles; if it is too high, then the reward will be too sparse.

Let $M = 10$ here.

Reset, Step, and Close

Before starting or when the game finishes, the environment needs to be reset. Reset needs to initialize all the attributes.

Function $step(action)$ takes in an *action* from the action set for the agent and updates the attributes such as state, and observations.

Function *close* is to close the game environment.

6.6 Experiment Result

In this section, we discuss the experimental results of applying the Deep Q-Network (DQN) algorithm and the Double DQN (DDQN) algorithm to the parking lot optimization problem. The discount rate is set to 0.99, considering the importance of the initial actions and their impact on the final rewards.

For DQN algorithm, reward and loss for different learning rate lr is shown in the following:

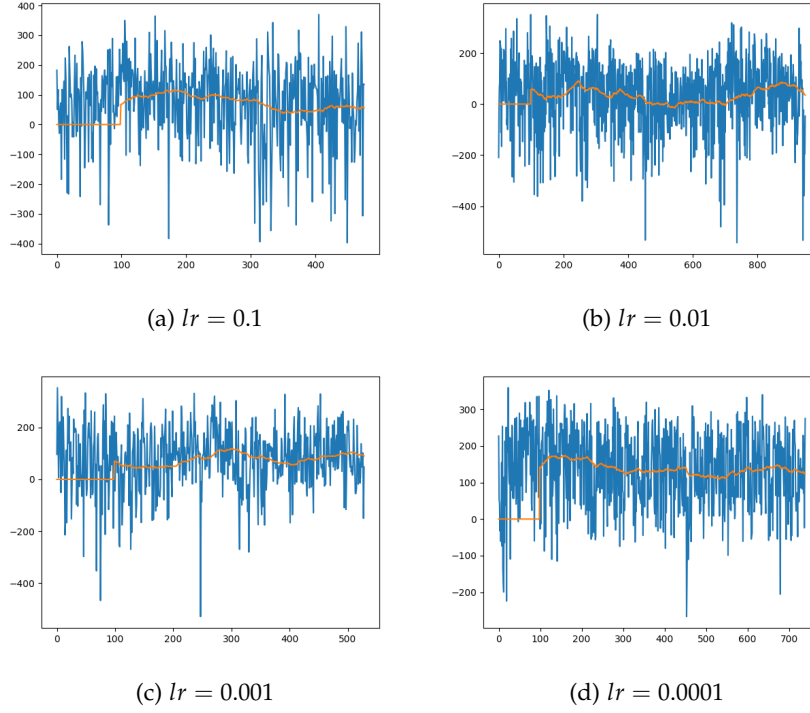


Figure 6.12: Reward for different lr using DQN Algorithm

The loss for DQN:

6 Experiment

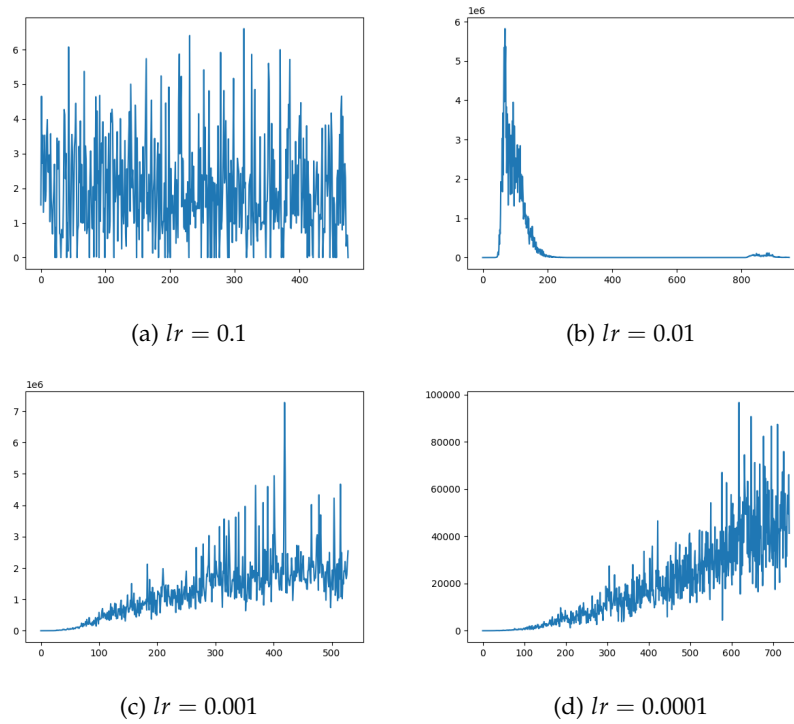
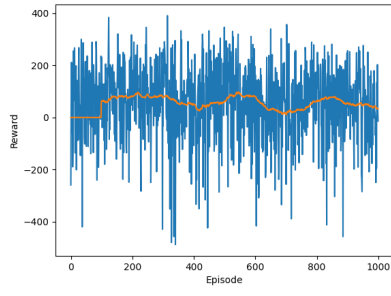


Figure 6.13: Loss for different lr using DQN Algorithm

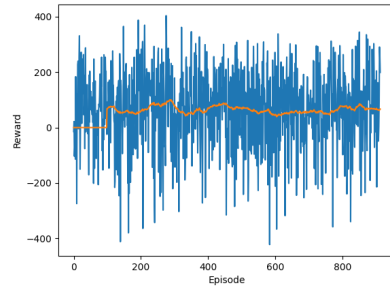
We observe that for $lr = 0.01$, although the loss exhibits a clear decrease and convergence, there is no significant improvement or clear increase in the reward. This suggests that the DQN algorithm may struggle to learn an effective policy that optimizes the parking process with this particular learning rate.

Next, it is the application of the DDQN algorithm:

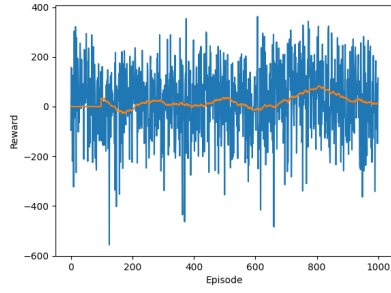
6 Experiment



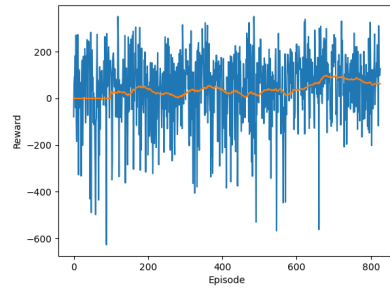
(a) $lr = 0.1$



(b) $lr = 0.01$



(c) $lr = 0.001$



(d) $lr = 0.0001$

Figure 6.14: Reward for different lr using DDQN Algorithm

The loss for DDQN:

6 Experiment

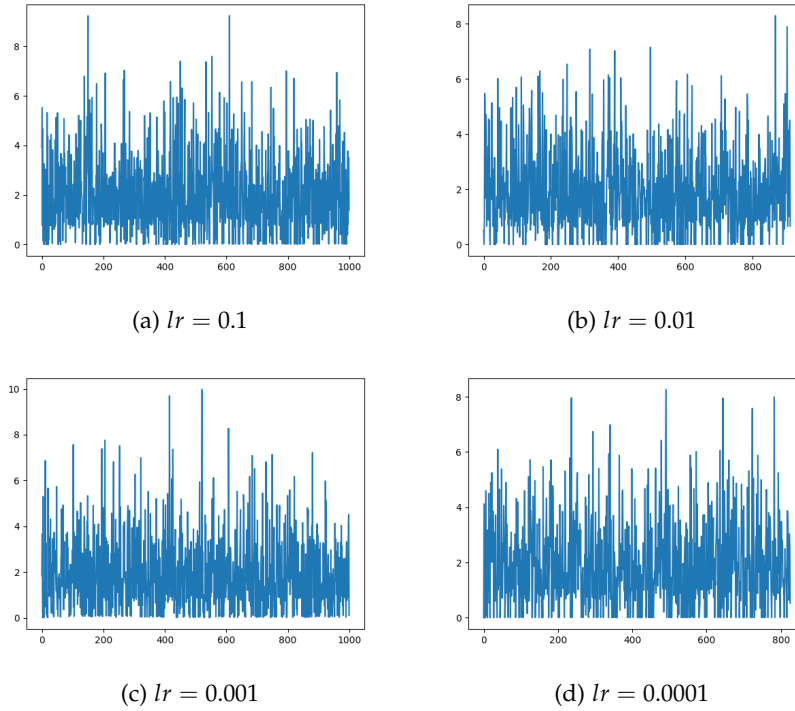


Figure 6.15: Loss for different lr using DDQN Algorithm

Unfortunately, it is observed that there is no clear convergence in the reward values across the different lr settings. This indicates that the DDQN algorithm also struggles to find an optimal solution and improve the parking process. Similar to the reward results, there is no clear convergence or stabilization in the loss values, further suggesting that the DDQN algorithm fails to effectively learn and optimize the parking allocation strategy.

The lack of clear convergence in both the reward and loss values for both the DQN and DDQN algorithms raises concerns about their effectiveness in addressing the parking lot optimization problem. These results indicate that the algorithms may encounter challenges in learning an optimal policy and improving the parking process performance.

Possible reasons for the limited success of the DQN and DDQN algorithms in this context could include:

- Complexity of the problem: There is clear delay between the action and reward. Besides, the environment is dynamic due to the movement of vehicles. This complexity may make it difficult for the algorithms to learn effective policies and optimize the parking allocation strategy.
- Insufficient exploration: The exploration-exploitation trade-off plays a crucial role in reinforcement learning algorithms. Since the state-action pairs set is pretty large and the train episodes numbers are no more than 2000 due to the time and computation limitation, it is possible that the DQN and DDQN algorithms fail to explore the solution space adequately, resulting in limited convergence.

- Hyperparameter selection: The choice of hyperparameters, such as the learning rate, discount rate, and batch size, decay rate, can significantly impact the performance of reinforcement learning algorithms. In this experiment, the selected lr values may not have been optimal for achieving convergence and optimal solutions.
- Network architecture and training process: The design of the neural network architecture and the training process used in the DQN and DDQN algorithms can influence their performance. In this experiment, it is possible that the chosen network architecture and training process were not well-suited for the parking lot optimization problem.

Overall, the results suggest that while the DQN and DDQN algorithms show promise in other domains, they may not be the most effective solutions for addressing the complexities of parking lot optimization.

Bibliography

[noa] NetLogo Web: patches.

- [2] Afrin, T. and Yodo, N. (2020). A Survey of Road Traffic Congestion Measures towards a Sustainable and Resilient Transportation System. *Sustainability*, 12(11):4660. Number: 11 Publisher: Multidisciplinary Digital Publishing Institute.
- [3] Anusha, T. and Pushpalatha, M. (2022). Efficient Communication Model for a Smart Parking System with Multiple Data Consumers. *Smart Cities*, 5(4):1536–1553.
- [4] Arellano-Verdejo, J., Alonso-Pecina, F., Alba, E., and Guzmán Arenas, A. (2019). Optimal allocation of public parking spots in a smart city: problem characterisation and first algorithms. *Journal of Experimental & Theoretical Artificial Intelligence*, 31(4):575–597. Publisher: Taylor & Francis .eprint: <https://doi.org/10.1080/0952813X.2019.1591522>.
- [5] Asaduzzaman, A., Chidella, K. K., and Mridha, M. F. (2015). A time and energy efficient parking system using Zigbee communication protocol. In *SoutheastCon 2015*, pages 1–5. ISSN: 1558-058X.
- [6] Athira, A., Lekshmi, S., Vijayan, P., and Kurian, B. (2019). Smart Parking System Based On Optical Character Recognition. In *2019 3rd International Conference on Trends in Electronics and Informatics (ICOEI)*, pages 1184–1188.
- [7] Bibi, N., Majid, M. N., Dawood, H., and Guo, P. (2017). Automatic Parking Space Detection System. In *2017 2nd International Conference on Multimedia and Image Processing (ICMIP)*, pages 11–15.
- [8] Bock, F., Di Martino, S., and Origlia, A. (2020). Smart Parking: Using a Crowd of Taxis to Sense On-Street Parking Space Availability. *IEEE Transactions on Intelligent Transportation Systems*, 21(2):496–508. Conference Name: IEEE Transactions on Intelligent Transportation Systems.
- [9] Busoniu, L., Babuska, R., and De Schutter, B. (2010). Multi-agent Reinforcement Learning: An Overview. In *Studies in Computational Intelligence*, volume 310, pages 183–221. Journal Abbreviation: Studies in Computational Intelligence.
- [10] Chen, Z., Xu, Z., Zangui, M., and Yin, Y. (2016). Analysis of Advanced Management of Curbside Parking. *Transportation Research Record*, 2567(1):57–66. Publisher: SAGE Publications Inc.
- [11] Digani, V., Sabbatini, L., Secchi, C., and Fantuzzi, C. (2014). Hierarchical traffic control for partially decentralized coordination of multi AGV systems in industrial environments. In *2014 IEEE International Conference on Robotics and Automation (ICRA)*, pages 6144–6149. ISSN: 1050-4729.

Bibliography

- [12] Dogaroglu, B., Caliskanelli, S. P., and Tanyel, S. (2021). Comparison of intelligent parking guidance system and conventional system with regard to capacity utilisation. *Sustainable Cities and Society*, 74:103152.
- [13] Duan, M., Wu, D., and Liu, H. (2020). Bi-level programming model for resource-shared parking lots allocation. *Transportation Letters*, 12(7):501–511. Publisher: Taylor & Francis .eprint: <https://doi.org/10.1080/19427867.2019.1631596>.
- [14] Fransen, K. J. C., van Eekelen, J. A. W. M., Pogromsky, A., Boon, M. A. A., and Adan, I. J. B. F. (2020). A dynamic path planning approach for dense, large, grid-based automated guided vehicle systems. *Computers & Operations Research*, 123:105046.
- [15] Geng, Y. and Cassandras, C. G. (2013). New “Smart Parking” System Based on Resource Allocation and Reservations. *IEEE Transactions on Intelligent Transportation Systems*, 14(3):1129–1139. Conference Name: IEEE Transactions on Intelligent Transportation Systems.
- [16] Henao, A. and Marshall, W. E. (2013). Parking at Sporting Event Stadiums in Denver, Colorado. *Transportation Research Record*, 2359(1):17–26. .eprint: <https://doi.org/10.3141/2359-03>.
- [17] Horng, G.-J. (2014). Using Cellular Automata for Parking Recommendations in Smart Environments. *PLOS ONE*, 9(8):e105973. Publisher: Public Library of Science.
- [18] Huang, C., Lu, R., Lin, X., and Shen, X. (2018). Secure Automated Valet Parking: A Privacy-Preserving Reservation Scheme for Autonomous Vehicles. *IEEE Transactions on Vehicular Technology*, 67(11):11169–11180. Conference Name: IEEE Transactions on Vehicular Technology.
- [19] Kumar, A. and Mudgal, A. (2023). Effect of vehicle size on crash risk in a heterogeneous traffic scenario: a bivariate extreme value approach. *Transportation Letters*, 0(0):1–16. Publisher: Taylor & Francis .eprint: <https://doi.org/10.1080/19427867.2023.2171843>.
- [20] Lin, Y.-Z. and Chen, W.-H. (2017). A simulation-based multiclass, multimodal traffic assignment model with departure time for evaluating traffic control plans of planned special events. *Transportation Research Procedia*, 25:1352–1379.
- [21] Lu, J., Li, B., Li, H., and Al-Barakani, A. (2021). Expansion of city scale, traffic modes, traffic congestion, and air pollution. *Cities*, 108:102974.
- [22] Mahendra, B. M., Sonoli, S., Bhat, N., Raju, and Raghu, T. (2017). IoT based sensor enabled smart car parking for advanced driver assistance system. In *2017 2nd IEEE International Conference on Recent Trends in Electronics, Information & Communication Technology (RTEICT)*, pages 2188–2193.
- [23] Makarova, I., Mavrin, V., Sadreev, D., Buyvol, P., Boyko, A., and Belyaev, E. (2022). Rational Organization of Urban Parking Using Microsimulation. *Infrastructures*, 7(10).
- [24] McGurrin, M., Vasudevan, M., and Tarnoff, P. (2012). *Benefits of Dynamic Mobility Applications: Preliminary Estimates from the Literature*.
- [25] Mladenović, M., Delot, T., Laporte, G., and Wilbaut, C. (2021). A scalable dynamic parking allocation framework. *Computers & Operations Research*, 125:105080.

- [26] Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A. A., Veness, J., Bellemare, M. G., Graves, A., Riedmiller, M., Fidjeland, A. K., Ostrovski, G., Petersen, S., Beattie, C., Sadik, A., Antonoglou, I., King, H., Kumaran, D., Wierstra, D., Legg, S., and Hassabis, D. (2015). Human-level control through deep reinforcement learning. *Nature*, 518(7540):529–533. Number: 7540 Publisher: Nature Publishing Group.
- [27] Nakazato, T., Fujimaki, Y., and Namerikawa, T. (2022). Parking Lot Allocation Using Rematching and Dynamic Parking Fee Design. *IEEE Transactions on Control of Network Systems*, 9(4):1692–1703. Conference Name: IEEE Transactions on Control of Network Systems.
- [28] Nawaz, S., Efstratiou, C., and Mascolo, C. (2013). ParkSense: a smartphone based sensing system for on-street parking. In *Proceedings of the 19th annual international conference on Mobile computing & networking*, MobiCom '13, pages 75–86, New York, NY, USA. Association for Computing Machinery.
- [29] Patil, B. K., Deshpande, A., Suryavanshi, S., Magdum, R., and Manjunath, B. (2018). Smart Parking System for Cars. In *2018 International Conference on Recent Innovations in Electrical, Electronics & Communication Engineering (ICRIEECE)*, pages 1118–1121.
- [30] Poumanyvong, P., Kaneko, S., and Dhakal, S. (2012). Impacts of urbanization on national transport and road energy use: Evidence from low, middle and high income countries. *Energy Policy*, 46:268–277.
- [31] Prabagar, A., Madhavaraja, N. S., Arunmozhi, S., and Manic, K. S. (2021). Artificial Vision Based Smart Urban Parking System. In *2021 International Conference on System, Computation, Automation and Networking (ICSCAN)*, pages 1–4.
- [32] Promy, N. and Islam, S. (2019). A Smart Android Based Parking System to Reduce the Traffic Congestion of Dhaka City. In *2019 21st International Conference on Advanced Communication Technology (ICACT)*, pages 124–128. ISSN: 1738-9445.
- [33] Rahman, M. M., Najaf, P., Fields, M. G., and Thill, J.-C. (2022a). Traffic congestion and its urban scale factors: Empirical evidence from American urban areas. *International Journal of Sustainable Transportation*, 16(5):406–421. Publisher: Taylor & Francis .eprint: <https://doi.org/10.1080/15568318.2021.1885085>.
- [34] Rahman, M. T., Zhang, Y., Arani, S. A., and Shao, W. (2022b). MDLpark: Available Parking Prediction for Smart Parking Through Mobile Deep Learning. In Ma, H., Wang, X., Cheng, L., Cui, L., Liu, L., and Zeng, A., editors, *Wireless Sensor Networks*, , pages 182–199, Singapore. Springer Nature.
- [35] Rajé, F., Tight, M., and Pope, F. D. (2018). Traffic pollution: A search for solutions for a city like Nairobi. *Cities*, 82:100–107.
- [36] Ruan, J.-M., Liu, B., Wei, H., Qu, Y., Zhu, N., and Zhou, X. (2016). How Many and Where to Locate Parking Lots? A Space-time Accessibility-Maximization Modeling Framework for Special Event Traffic Management. *Urban Rail Transit*, 2(2):59–70.
- [37] Sangeetha, N., Amutha, S., Deepa, A. R., and Joyal, S. (2022). IoT based Smart Parking Assistant for Smart Cities. In *2022 International Conference on Edge Computing and Applications (ICECAA)*, pages 593–598.

- [38] Sayarshad, H. (2023). Designing intelligent public parking locations for autonomous vehicles. *Expert Systems with Applications*, 222:119810.
- [39] Shao, C., Zhao, Y., Yu, L., and Xu, J. (2008). Case Analysis on Vehicle Parking and Departure Plans for the National Stadium. *Journal of Transportation Systems Engineering and Information Technology*, 8(6):52–55.
- [40] Shin, J.-H. and Jun, H.-B. (2014). A study on smart parking guidance algorithm. *Transportation Research Part C: Emerging Technologies*, 44:299–317.
- [41] Shin, J.-H., Jun, H.-B., and Kim, J.-G. (2018). Dynamic control of intelligent parking guidance using neural network predictive control. *Computers & Industrial Engineering*, 120:15–30.
- [42] Su, Y., Liu, X., and Li, X. (2020). Research on Traffic Congestion Based on System Dynamics: The Case of Chongqing, China. *Complexity*, 2020:6123896. Publisher: Hindawi.
- [43] Sui, X., Ye, X., Wang, T., Yan, X., Chen, J., and Ran, B. (2022). Microscopic Simulating the Impact of Cruising for Parking on Traffic Efficiency and Emission with Parking-and-Visit Test Data. *International Journal of Environmental Research and Public Health*, 19(15):9127. Number: 15 Publisher: Multidisciplinary Digital Publishing Institute.
- [44] Sutton, R. S. and Barto, A. G. (2018). *Reinforcement Learning, second edition: An Introduction*. MIT Press. Google-Books-ID: uWV0DwAAQBAJ.
- [45] Tempelmeier, N., Dietze, S., and Demidova, E. (2020). Crosstown traffic - supervised prediction of impact of planned special events on urban traffic. *GeoInformatica*, 24(2):339–370.
- [46] Viard, V. B. and Fu, S. (2015). The effect of Beijing’s driving restrictions on pollution and economic activity. *Journal of Public Economics*, 125:98–115.
- [47] Vo, T. T. A., van der Waerden, P., and Wets, G. (2016). Micro-simulation of Car Drivers’ Movements at Parking Lots. *Procedia Engineering*, 142:100–107.
- [48] Wang, Z., Ahmed, Z., Zhang, B., and Wang, B. (2019). The nexus between urbanization, road infrastructure, and transport energy demand: empirical evidence from Pakistan. *Environmental Science and Pollution Research*, 26(34):34884–34895.
- [49] Watkins, C. J. C. H. and Dayan, P. (1992). Q-learning. *Machine Learning*, 8(3):279–292.
- [50] Wu, D., Zeng, Z., Shi, F., Yu, W., Wu, T., and Liu, Q. (2022). Human as a Service: Towards Resilient Parking Search System With Sensorless Sensing. *IEEE Transactions on Intelligent Transportation Systems*, 23(8):13863–13877. Conference Name: IEEE Transactions on Intelligent Transportation Systems.
- [51] Xie, J., He, Z., and Zhu, Y. (2022). A DRL based cooperative approach for parking space allocation in an automated valet parking system. *Applied Intelligence*.
- [52] Xie, R., Wei, D., Han, F., Lu, Y., Fang, J., Liu, Y., and Wang, J. (2019). The effect of traffic density on smog pollution: Evidence from Chinese cities. *Technological Forecasting and Social Change*, 144:421–427.
- [53] Xie, Z., Wu, X., Guo, J., and Zhan, Z. (2021). Parking lot allocation model considering conversion between dynamic and static traffic. *Journal of Intelligent & Fuzzy Systems*, 41(4):5207–5217. Publisher: IOS Press.

Bibliography

- [54] Xu, D. and Sun, H. (2022). Research on Parking Service Optimization Based on Permit Reservation and Allocation. *Journal of Advanced Transportation*, 2022.
- [55] Xue, Z., Cheng, J., and Ji, K. (2021). An Allocation Model of Shared Parking with Continuous Time Windows. In *2021 IEEE 8th International Conference on Industrial Engineering and Applications (ICIEA)*, pages 616–620.
- [56] Yang, G., Chen, J., Lu, K., and Zhang, C. (2021). Improving Utilization Efficiency and Drivers' Degree of Satisfaction in Urban Complex Parking Lot through Modelling the Dynamic Allocation of Parking Spaces. *Applied Sciences*, 11(18):8680. Number: 18 Publisher: Multidisciplinary Digital Publishing Institute.
- [57] Zhang, J., Li, Z., Li, L., Li, Y., and Dong, H. (2021). A bi-level cooperative operation approach for AGV based automated valet parking. *Transportation Research Part C: Emerging Technologies*, 128.
- [58] Zhao, C., Li, S., Wang, W., Li, X., and Du, Y. (2018). Advanced Parking Space Management Strategy Design: An Agent-Based Simulation Optimization Approach. *Transportation Research Record: Journal of the Transportation Research Board*, 2672:036119811875867.
- [59] Zhao, C., Liao, F., Li, X., and Du, Y. (2021). Macroscopic modeling and dynamic control of on-street cruising-for-parking of autonomous vehicles in a multi-region urban road network. *Transportation Research Part C: Emerging Technologies*, 128:103176.
- [60] Zhao, Z., Zhang, Y., Ji, K., and Qi, H. (2020). Neural-Network-Based Dynamic Distribution Model of Parking Space Under Sharing and Non-Sharing Modes. *Sustainability*, 12(12):4864. Number: 12 Publisher: Multidisciplinary Digital Publishing Institute.